



PRODUCT
DATASHEET '24

ADVANCE ANALYTIK

REVOLUTIONIZING ONLINE MONITORING SOLUTIONS



OPTICS 1000 - Total Hardness

TOTAL HARDNESS (Hd)

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Optics 1000 Series

TOTAL HARDNESS (Hd)

Method - Colorimetric

After adding the reagents to establish the desirable conditions to the sample, the volume of titrant is charged.

Thanks to the module syringe driver minimum dose (0.015 ml), the accuracy achieved is really high, with 3 drops differences as much between 2 consecutives analysis of the same standard. Moreover, the software has been improved to speed up the initial dose, whereas when the inflection point is close, the titrant is added slowly, to achieve precision.



Principle of measurement

The calcium and magnesium complexes with the Erichrome Black-T to give a blue-violet colored complex. This, when titrated by the addition of EDTA, is broken up as the calcium EDTA/ Magnesium ESTA complexes are more stable and when excess EDTA is present the Erichrome Black -T-EDTA complex is formed which is a rose colored complex.



Advantages of the method

The complexometric titration of both the calcium and magnesium, which are the main contributors to the hardness of water, gives a very sharp end point because at the end-point neither the Ca-EBT nor the EDTA-EBT complexes exist.

Specifications

RANGE	From 0 to 100 ppm /250 ppm/ 500ppm / 1000 ppm Adjustable to lower/higher concentrations
ACCURACY	±2%
ANALYSIS TIME	around 20 minutes
REPEATABILITY	±2%
CALIBRATION	one point
LED WAVELENGTH	650 nm

Reagents consumption

- Reagent 1: 0.37 ml / analysis - 0.3L / month
- Reagent 2: 0.37 ml / analysis - 0.3L / month
- Reagent 3: 0.37 ml / analysis - 0.3L / month

Monthly consumption calculated assuming 1 analysis per hour



**Note -**

This data sheet serves as general information about the Optics 1000 - Total Hardness (Hd). For specific technical details, installation guidelines, and troubleshooting assistance, please refer to the official user manual provided with the product.

For inquiries and detailed technical information, please contact sales@advanceanalytik.com.



LET'S CONNECT !

Let's work together to find a solution that works for you



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